

Statistical Modelling of Equity Volatility

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August 2026

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1 Introduction

Volatility is an important measure of uncertainty in financial markets and plays a central role in areas such as risk management and derivatives pricing. Being able to forecast changes in volatility can provide useful information when assessing future market conditions.

This project investigates how effectively different statistical models can forecast short-term equity volatility. Using daily market data from five large US equities — Apple, Microsoft, Amazon, Nvidia and Alphabet (Google) — I calculate 20-day rolling volatility and use it as the forecasting target. Four approaches of increasing complexity are compared: a naïve forecast, a moving-average model, a linear regression model and a GARCH model.

The models are evaluated using out-of-sample predictions, with their performance compared using measures including mean absolute error and root mean squared error. The aim is to determine which model produces the most accurate forecasts, investigate why simpler models may perform competitively and how the characteristics of the data affect model performance.

2 Data and Volatility Measurement

2.1 Data

The project uses daily market data for five large US equities: Apple (AAPL), Microsoft (MSFT), Amazon (AMZN), Nvidia (NVDA) and Alphabet (GOOGL). The data was obtained through the Alpha Vantage API and contains the daily opening price, highest traded price, lowest traded price, closing price and trading volume.

For each asset, one observation corresponds to a single trading day. The closing price is used as the primary measure of the value of the asset at the end of each trading day. Trading volume records the total number of shares traded during the day.

The data is ordered chronologically and converted into numerical and datetime formats before analysis. The dataset is then divided chronologically into training and test periods, with the final 20% of observations reserved for out-of-sample evaluation.

2.2 Daily Returns

Rather than modelling the raw share price directly, daily percentage returns are used. For a security with closing price (P_t) on day (t), the daily return is defined as

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}.$$

This measures the proportional change in the closing price from one trading day to the next, which is more suitable for statistical analysis of volatility.

2.3 Measuring Volatility

Volatility is measured using the rolling standard deviation of daily returns over a 20-day window. For day (t), the volatility measure is

$$\sigma_t = \sqrt{\frac{1}{19} \sum_{i=0}^{19} (R_{t-i} - \bar{R}_t)^2},$$

where

$$\bar{R}_t = \frac{1}{20} \sum_{i=0}^{19} R_{t-i}$$

is the mean return over the same 20-day period.

A 20-day window was chosen to provide a relatively short-term measure of market volatility while still using enough observations to produce a meaningful estimate. The four models then use this volatility series as their forecasting target

Since the volatility is calculated over rolling 20-day windows, consecutive observations share 19 of the same returns. This makes the volatility series relatively persistent, which is important when interpreting the model results.

3 Forecasting Models

Four models are used to forecast next-day equity volatility. They range from simple methods to a more advanced time-series model. Comparing these models allows us to see whether more complex methods produce better forecasts.

3.1 Naïve Model

The first model provides a simple benchmark against which the other models can be assessed. It assumes that the most recent observed volatility is the best estimate of the following day's volatility. Therefore,

$$\hat{\sigma}_{t+1} = \sigma_t,$$

where σ_t is the observed 20-day volatility on day (t), and $\hat{\sigma}_{t+1}$ is the forecast for day ($t+1$).

The model requires no parameter estimation or training. Despite its simplicity, it provides a useful benchmark because volatility tends to be persistent over short periods. The 20-day rolling volatility also means that consecutive observations overlap significantly, as discussed in Section 2.3.

3.2 Moving Average Model

The second model forecasts future volatility using the recent history of the volatility series. A five-day moving average is used, giving

$$\frac{1}{5} \sum_{i=0}^4 \sigma_{t-i}.$$

This model smooths short-term fluctuations by averaging the most recent five volatility observations. The model uses recent volatility to predict future volatility while smoothing out unusually high or low values.

The five-day window was selected as a simple short-term forecasting horizon. Unlike the naïve model, which uses only the latest volatility, the moving average uses several recent observations.

3.3 Linear Regression Model

The third model uses linear regression to test whether additional market information can improve volatility forecasts. It uses three variables based on information available before the forecast date:

$$X_1 = \sigma_{t-1},$$

$$X_2 = |R_{t-1}|,$$

and

$$X_3 = \frac{V_{t-1}}{\bar{V}_{t-1}},$$

where σ_{t-1} is the previous day's volatility, $|R_{t-1}|$ is the absolute return from the previous day, V_{t-1} is the previous day's trading volume, and $\bar{V}_{t-1}$ is its recent average volume.

The forecasting equation is therefore

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \epsilon_t.$$

The coefficients are estimated using the training data and then used to make predictions on the test data. This tests whether information beyond recent volatility improves the forecasts.

3.4 GARCH Model

The fourth model uses a Generalised Autoregressive Conditional Heteroskedasticity (GARCH) model, which is specifically designed to model time-varying volatility in financial returns.

A GARCH(1,1) model specifies the conditional variance as

$$\omega + \alpha\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2,$$

where ω is a constant term, ϵ_{t-1}^2 represents the squared return shock from the previous period, and σ_{t-1}^2 is the previous conditional variance. The parameters ω , α and β are estimated from the training data.

The model accounts for two features: the effect of recent price changes on volatility and the tendency for volatility to persist over time. Unlike the first three models, GARCH models how volatility changes over time rather than simply using or averaging recent volatility.

The GARCH forecasts are compared with the same test data as the other models. This allows their performance to be compared fairly, while recognising that GARCH measures volatility differently from the rolling volatility used as the target.

4 Evaluation Methodology

The four models are tested using the final 20% of the data. The data is ordered by date, with the first 80% used for training and the remaining 20% used for testing. This prevents future information from affecting the forecasts and ensures the models are compared fairly.

For each day in the test period, the predicted volatility is compared with the observed 20-day rolling volatility. Mean Absolute Error (MAE) is used to compare the forecasting accuracy of the four models.

4.1 Mean Absolute Error

The Mean Absolute Error measures the average absolute difference between the predicted and observed volatility:

$$\frac{1}{n} \sum_{t=1}^n |\hat{\sigma}_t - \sigma_t|,$$

where $\hat{\sigma}_t$ is the predicted volatility, σ_t is the observed volatility and n is the number of test observations.

MAE provides a direct measure of the typical size of a model's forecasting error. A lower MAE indicates that the model's forecasts are on average closer to the observed volatility.

4.2 Additional Analysis

Forecast accuracy is not measured using one error measure alone. The predictions are also compared with the observed volatility over time to see when

models perform well or poorly. Model performance is also compared across assets, as a model that performs well for one stock may not perform as well for another.

The results are analysed at both the individual asset and overall levels. This allows us to determine whether one forecasting approach consistently outperforms the others, or whether model performance depends on the characteristics of the asset and the behaviour of its volatility.

5 Results

The forecasting models were evaluated on the final (20%) of observations for each security. Mean Absolute Error (MAE) was used as the primary measure of forecast accuracy, with lower values indicating that predicted volatility was closer to the observed value.

5.1 Overall Model Performance

Table 1 shows the MAE obtained by each model for the five securities. The results show a clear advantage for the simpler forecasting approaches. The naïve model achieves the lowest MAE for four of the five assets, while the linear regression model performs best for NVDA. The GARCH model has the highest MAE for every asset.

Table 1: Mean Absolute Error of each volatility forecasting model.

Stock	Naïve	Moving Average	Regression	GARCH	Best Model
AAPL	0.0006	0.0015	0.0009	0.0024	Naïve
AMZN	0.0010	0.0022	0.0026	0.0133	Naïve
GOOGL	0.0010	0.0020	0.0017	0.0033	Naïve
MSFT	0.0009	0.0022	0.0024	0.0045	Naïve
NVDA	0.0009	0.0016	0.0007	0.0016	Regression

The naïve model performs particularly well. Although it is the simplest approach, using the latest observed volatility gives the lowest MAE for four of the five assets. The moving-average model performs worse than the naïve model, while the regression model gives mixed results, only performing best for NVDA. The GARCH model performs substantially worse than the other approaches across all five assets. Its error is particularly large for AMZN. This suggests that the added complexity of the GARCH model did not lead to more accurate forecasts for the data used in this study.

5.2 Performance Across Assets

The results also show that model performance varies between assets. The naïve model performs particularly well for AAPL and MSFT, while the regression

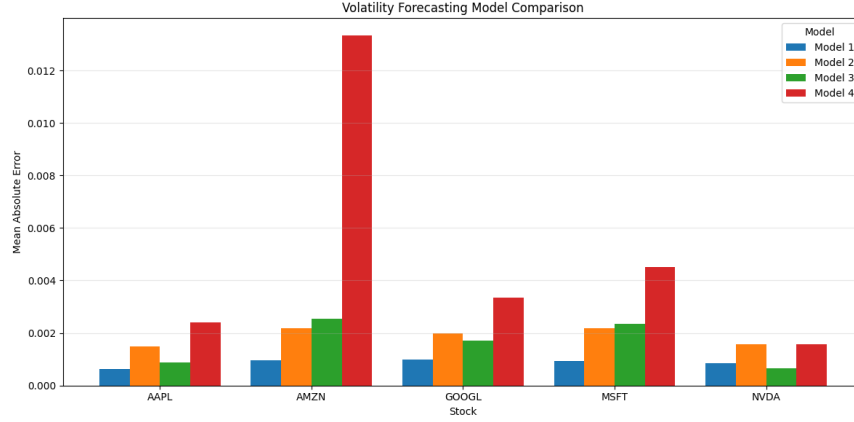


Figure 1: Mean Absolute Error of the four volatility forecasting models across the five assets.

model gives the lowest error for NVDA. This suggests that there may not be a single forecasting approach that is optimal across all assets.

For NVDA, the regression model achieves the lowest MAE. This suggests that information such as the previous day’s absolute return and relative trading volume may help forecast its volatility.

However, this advantage is not observed across the other assets. For AAPL, AMZN, GOOGL and MSFT, the additional variables in the regression model did not improve the forecasts compared with simply using the most recent volatility.

5.3 Interpretation of the Results

The strong performance of the naïve model can be explained by how volatility is calculated in this study. Since volatility is based on a 20-day rolling standard deviation, consecutive observations share 19 of their 20 returns. This makes the volatility series highly persistent, meaning that today’s volatility is likely to be similar to tomorrow’s and making the simple naïve model difficult to outperform.

The moving-average model also benefits from this persistence by using several recent volatility observations. However, averaging can make the forecast slower to respond to sudden changes in volatility, which may explain why it generally performs worse than the naïve model.

The regression model uses additional market information to make its forecasts. Its mixed performance suggests that recent returns and relative trading volume may help predict volatility for some assets, but their usefulness is not consistent across all five assets.

Finally, the GARCH model’s poor performance should be interpreted carefully. The dataset provides relatively few observations for estimating its parameters, and GARCH predicts conditional volatility rather than the 20-day

rolling volatility used as the target. These differences may have contributed to its weaker performance.

6 Discussion

6.1 Overall Model Performance

The results show that the simplest forecasting approach performed surprisingly well. The naïve model achieved the lowest Mean Absolute Error (MAE) for four of the five securities, with the linear regression model performing best only for NVDA. This indicates that increasing model complexity did not consistently improve out-of-sample forecasting accuracy.

This result is notable because the naïve model does not estimate any parameters or use additional market information. It simply assumes that the most recent volatility is a good estimate of the following day’s volatility. Its strong performance suggests that the volatility measure is highly persistent over short periods.

6.2 Persistence of the Volatility Measure

A possible mathematical explanation for the performance of the naïve model comes from the construction of the volatility measure. Volatility is calculated as the sample standard deviation of daily returns over a window of 20 observations:

$$\sigma_t = \sqrt{\frac{1}{19} \sum_{i=0}^{19} (R_{t-i} - \bar{R}_t)^2}.$$

The following day’s volatility is calculated using the next rolling window,

$$\sigma_{t+1} = \sqrt{\frac{1}{19} \sum_{i=0}^{19} (R_{t+1-i} - \bar{R}_{t+1})^2}$$

These two windows share 19 of the same 20 daily returns, so consecutive volatility values are closely related. Unless the new return is very different from the previous returns, the volatility is unlikely to change substantially. This provides a natural explanation for why the assumption

$$\hat{\sigma}_{t+1} = \sigma_t$$

can perform well.

6.3 Performance of the Moving Average Model

The moving-average model performed worse than the naïve model for all five assets. Averaging recent volatility smooths out short-term changes, but this can make the forecast slower to react to sudden changes. For example, after a sharp rise in volatility, the naïve model immediately uses the new value, while the moving average is still influenced by the four previous observations.

The results suggest that, for the dataset considered, the most recent volatility observation contains more useful information about the following day’s volatility than a smoothed average of recent volatility.

6.4 Performance of the Regression Model

The linear regression model produced mixed results. It achieved the lowest MAE for NVDA, but was outperformed by the naïve model for the other four securities. The regression model uses lagged volatility, the previous day’s absolute return and relative trading volume. Its better performance for NVDA suggests that these variables may provide useful information about future volatility that is not captured by recent volatility alone. However, this was not the case for the other assets, suggesting that the usefulness of these variables varies between stocks.

The NVDA result should be interpreted carefully. This analysis cannot determine why regression performed better for NVDA, and more data across a larger number of assets would be needed to see whether this result is consistent or specific to this sample.

6.5 Performance of the GARCH Model

The GARCH model had the highest MAE for all five assets, with particularly poor performance for AMZN. This suggests that in this study, the added complexity of GARCH did not improve the volatility forecasts. However, this does not mean that GARCH models are generally worse, as the data and model setup may have affected its performance.

Firstly, the relatively small dataset provides limited observations for estimating the parameters of the GARCH model. After calculating returns and the 20-day rolling volatility measure, the number of available observations is reduced further, and only the first (80%) of these observations are used for model estimation. A GARCH(1,1) model estimates parameters such as

$$\omega, \quad \alpha, \quad \beta,$$

and reliable estimation becomes more difficult when the available sample is small.

Under standard assumptions, the uncertainty associated with parameter estimates generally decreases with sample size, roughly proportional to

$$\frac{1}{\sqrt{n}},$$

where n is the number of observations. Increasing the size of the training sample should provide more information with which to estimate the underlying volatility.

Secondly, the target variable in this project is 20-day rolling volatility, while GARCH forecasts conditional volatility based on past returns. Although these measures are related, they are not the same, which may have contributed to the weaker GARCH results.

6.6 Effect of Model Complexity

An important finding is that greater model complexity did not lead to more accurate forecasts. Although the models became more complex from the naïve model to the moving average, regression and GARCH models, their performance did not improve in the same way. This suggests that more advanced models are not always better, particularly when the target is already highly persistent. It also highlights the importance of using a simple benchmark to determine whether a more complex model actually provides an improvement.

6.7 Differences Between Assets

The naïve model performed best for AAPL, AMZN, GOOGL and MSFT, whereas the regression model performed best for NVDA.

These results suggest that no single model was consistently best across all five assets. Differences in volatility and trading behaviour may affect how well each model performs. However, only five assets were studied, so these findings should not be generalised to the wider stock market.

6.8 Limitations and Future Work

Several limitations should be considered when interpreting the results. The dataset contains only a small number of assets and a limited period of historical data, which may affect the reliability of the comparison, particularly for models such as GARCH that require more parameter estimation.

The choice of a 20-day rolling standard deviation also affects the results. Since consecutive windows overlap heavily, the volatility series is highly persistent, which may favour simpler models. Using a different measure of volatility could therefore lead to different results.

The analysis also uses a single chronological training-test split. While this prevents future data from affecting earlier forecasts, using rolling or expanding test periods would give a more reliable comparison of model performance.

Future work could use several years of data and a larger number of assets. Different measures of volatility and GARCH models could also be tested. A larger out-of-sample analysis would help determine whether the poor GARCH performance was mainly due to the limited data and choice of target, or whether it would remain with a larger dataset.

7 Conclusion

This project compared four statistical models for forecasting short-term equity volatility across five US equities. The naïve model performed best for four of the five assets, while linear regression performed best for NVDA. Overall, the more complex models did not consistently produce more accurate forecasts, with GARCH performing worst under the conditions of this study.

The strong performance of the simpler models can be partly explained by the persistence created by the 20-day rolling volatility measure. Overall, the results show that more complex models do not always produce better forecasts, highlighting the importance of simple benchmark models. Using more data, different volatility measures and other forecasting methods would help test these findings further.

Thank you for taking the time to read this report.